

2+ years work experience (U.S./International); 4+ years research/programming experience; 302 stars on GitHub.

EDUCATION

Ph.D. Student in AI for Geosciences University of Texas at Austin 2024 – Expected: May 2029
Reviewer for *ACM Computing Surveys*, *ACM AgentSkills*, *ACM TOSN*, and *PSS*
B.S. in Geosciences (Honors Class) University of Science and Technology of China 2020 – 2024
Finalist for the Guo Moruo Scholarship (USTC's highest undergraduate honor)
Founded one of the largest student clubs, grew from 0 to 1,300+ members. [News]

WORK EXPERIENCE

Lead Developer, ESM-bench: AI Agent Benchmark for Earth System Models UT-Austin | Dec. 2025 – Present

- Designed a 243-task benchmark across 3 Earth System Models (Noah-MP, CLM5, MOM6) evaluating whether LLM agents can bridge the physics-and-code understanding gap in 200k+ line Fortran/C codebases.
- Built a multi-model evaluation framework testing 5+ LLMs (GPT-5.5, Claude Opus, Gemini, etc) with 3 progressive hint levels and a 6-item physics-aware semantic and result-based rubric.
- Achieved <20% exact match across all frontier models, demonstrating that physical correctness in scientific code remains an unsolved challenge for current LLMs. [Preprint]

Co-Lead, FrontierShiftBench

ScienceIntelligence.org | Jun. 2026 – Present

- Co-lead an AI-science benchmark spanning dry computational tasks and wet-lab planning tasks.
- Drafted the task schema, contribution guide, public/hidden split, validators, and scoring rubrics.
- Built the expert-review workflow for auditing agent-generated tasks before format freeze.

Program Committee, Proposed NeurIPS 2026 Workshop

May 2026 – Present

- Support proposal and program review for [The Future of the AI Research Ecosystem](#), focused on benchmarks, datasets, evaluation protocols, artifact standards, and AI-scientist methods.
- Shape workshop topics on publication formats, citation graphs, review workflows, archives, and agent-verifiable research infrastructure.

Contributor, Agent-AI4Sci-Bench: Project-Level AI-for-Science Agent Benchmark

Apr. 2026 – Present

- Contributed to task development and review for a 70-task, 12-domain AI-for-science agent benchmark.
- Improved task quality and evaluation tooling: reference generators, custom scorers, preflight/GT checks, sandbox fixes, provenance, batch records, and reporting exports.

Co-Lead & Maintainer

ResearchSkills.ai | ScienceIntelligence.org | Remote | Mar. 2026 – Present

- Improved a one-command extraction flow that makes contributing tacit research expertise as easy as running Claude/Codex on your own research history and approving the distilled skills. [Site] [Org]
- Improved the contributor-draft/admin-approve workflow that routes approved skills to the public ScienceIntelligence/ResearchSkills registry through authenticated review.

AI Intern, AI Agent Knowledge Base Evaluation

19Pine.AI (Singapore) | Remote | Jul. 2025 – Aug. 2025

- Implemented a multi-dimensional evaluation system from scratch for PineAgent's RAG knowledge base.
- Developed a data sanitization module removing 2,000+ PII entries and noise from 1,000+ call sessions.
- Extracted Q&A pairs (knowledge/method/strategy) from call sessions to form the evaluation dataset.
- Deployed a concurrent LLM-as-judge + rule-based engine, measuring P/R/F1 to enable RLHF optimization.

CEO-Invited AI Consultant

Keshi Edu | Remote, Beijing | May 2025 – Jun. 2025

- Designed an architecture for a RAG-based knowledge base to customize graduate program applications.
- Identified critical bugs and exposed API keys, saving thousands of dollars in potential losses.

Research Assistant, ML Model/Multi-agent Development

UT-Austin | Austin, TX | Aug. 2025 – Jan. 2026

Project Lead, High-Performance Computing Allocation

NSF NCAR | Remote | Aug. 2025 – Present

- Proposed and secured NSF NCAR's 1k GPU + 22k CPU hours high-performance computing allocation.
- Integrating machine learning parameter calibration for a SOTA physics-based land surface model (Noah-MP).
- Developing a multi-expert AI agent system for automated parameterization for physics-based climate models.

PUBLICATIONS

- Mai, G., Lao, N., Zhang, J., Mao, L., Wang, Z., Wu, N., Janowicz, K., **Wu, K.**, Rao, J., Gao, S., & Zhu, R. (2026). On the Ethics of Generative GeoAI: Explainability, Bias, Hallucination, Accountability, Privacy, and Trust. In *Geography According to Foundation Models*, Vol. 422, pp. 215–232. IOS Press. [\[Chapter\]](#)
- **Wu, K.**, Cao, Y., & Mai, G. (2026). ESM-bench: A Benchmark for Evaluating Whether AI Agents Understand Earth System Model Physics and Code. (In prep to NeurIPS dataset). [\[Preprint\]](#)
- **Wu, K.**. Noah-Agent: A Multi-Expert AI Agent Framework for Automated Parameterization and Validation of Large-Scale Fortran Climate Models. (In prep). [\[Record\]](#)
- **Wu, K.**, Yi, W.*, Xue, X.*, Reid, I., & Lu, M. (2024). Diurnal and Seasonal Variations of Meteor Speed and Arrival Angle Observed by Mengcheng Meteor Radar. *JGR: Space Physics*. [\[Paper\]](#) [\[Code\]](#)

SKILLS

- Python, FastAPI, JavaScript, TypeScript, React, SQL, PyTorch, LangChain, RAG
- Claude Code, Codex, Openclaw, Hermes, Cursor, Docker, AWS/GCP/Azure, Redis, Git, HuggingFace